May, 24.

Day-40

Simple Network Management Protocol

- SNMP is an industry-standard framework and protocol that was originally released in 1988.

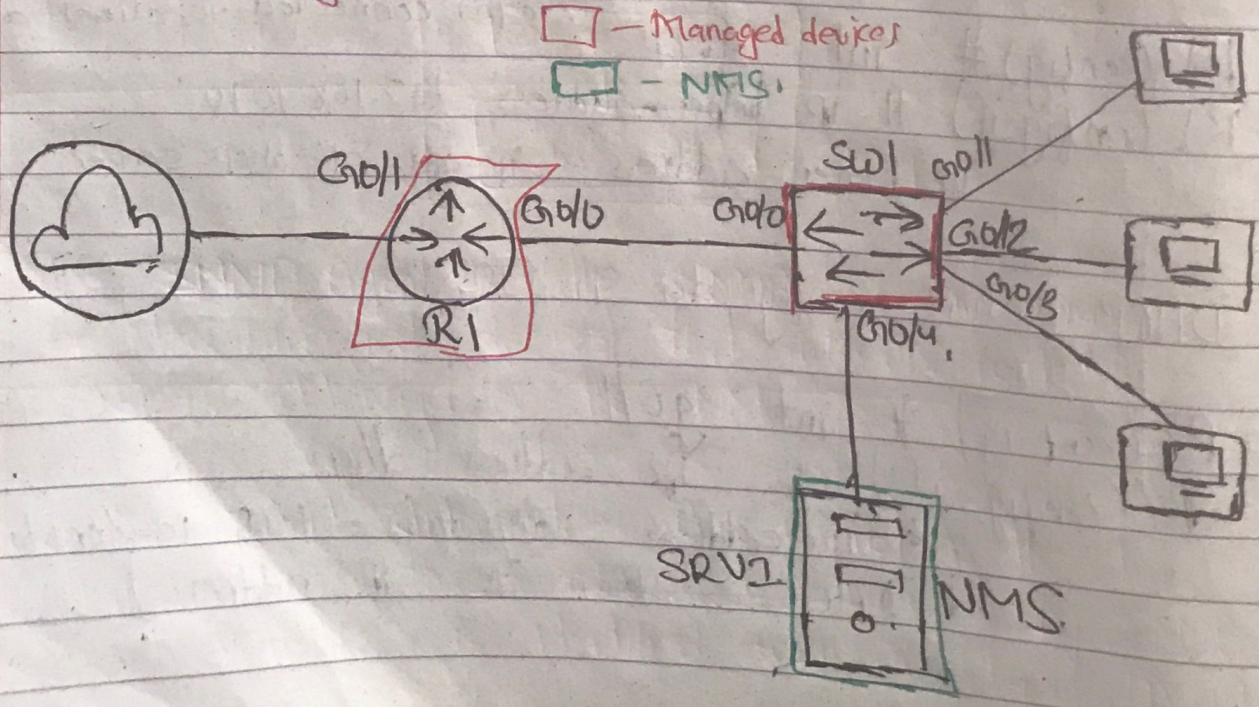
- Don't let the 'Simple' in the name fool you!
- SNMP can be used to monitor the status of devices, make configuration changes, etc.

- There are two main types of devices in SNMP:

1) Managed devices
   - These are devices being managed using SNMP.
   - For e.g: network devices like routers and switches.

2) Network Management Station (NMS)
   - The device/devices managing the managed devices.
   - This is the 'SNMP' server.

Network Diagram.



SNMP Operations

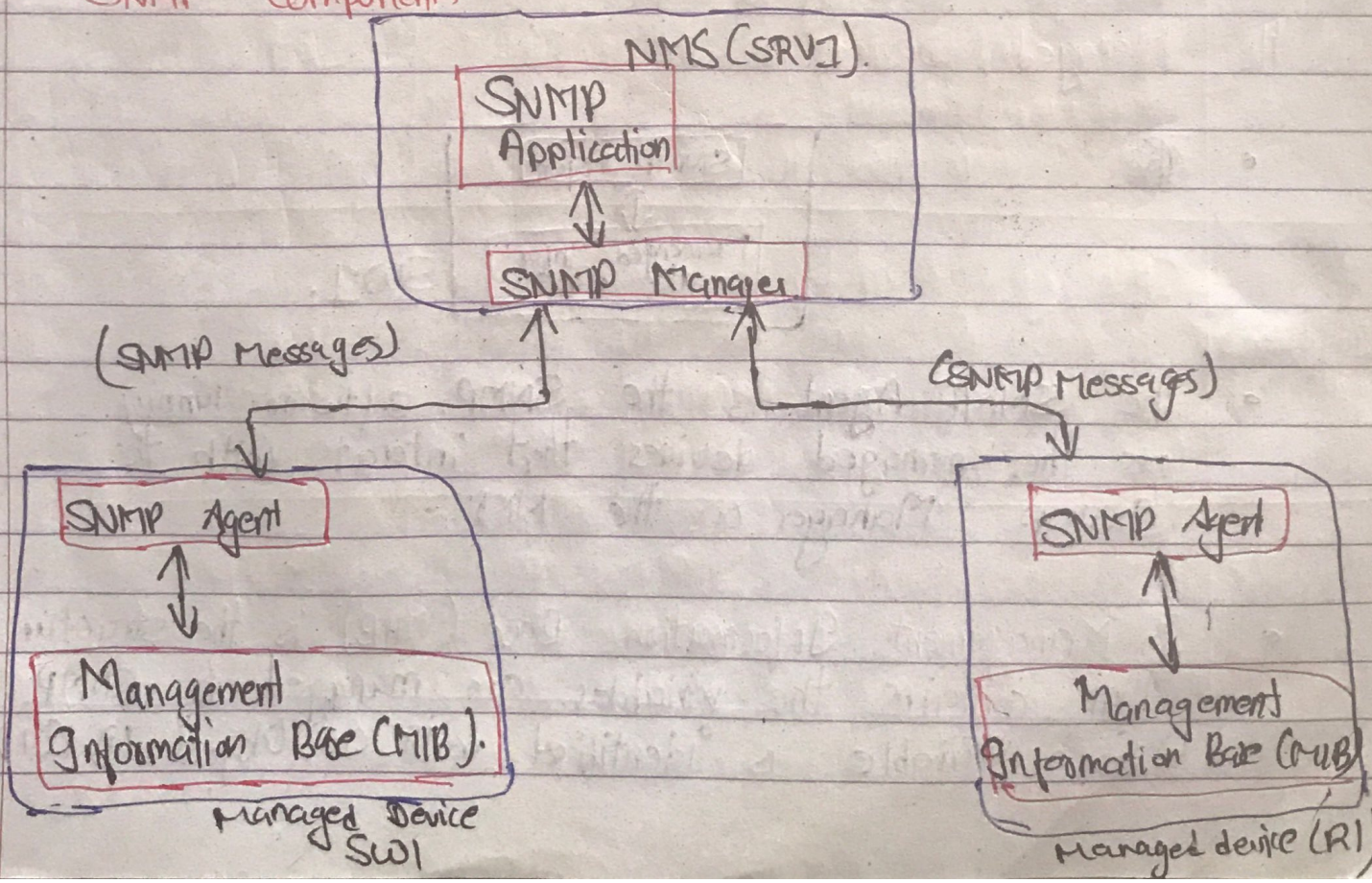
There are three main operations used in SNMP.

1) Managed devices can notify the NMS of events.
   Eg: SW1 G0/1 int had hardware failure & went down, SW1 can send NMS msg telling its G0/1 has gone down

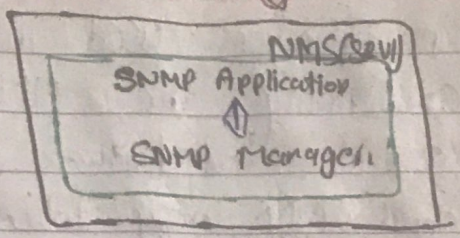
2) The NMS can ask the managed devices for information about their current status
   Eg: NMS, SRV1 asks R1 what its current CPU usage. usage is. R1 will then reply to SRV about its CPU usage

3) The NMS can tell the managed devices to change aspects of their configuration.
   For e.g. Say R1's IP add. of R1's G0/1 interface is 203.0.113.1/30. The NMS can tell R1 to change that IP add. to 203.0.113.5/30

SNMP Components



NMS (Network Management Station).



- SNMP Software on NMS

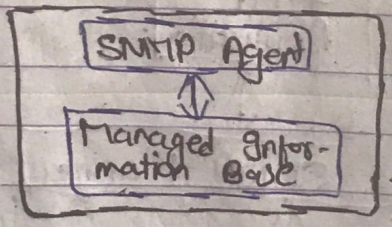
NMS is probably ain't a machine dedicated to SNMP. It could just be the network admin's PC which is running SNMP software.

- SNMP Manager :- is the software on the NMS that interacts with the managed devices. It receives notifications, send requests for information, sends configuration changes etc.

- SNMP Application - provides an interface for the network admin to interact with.
  - ↳ Displays alerts, statistics, charts etc.

Managed Devices

- [redacted]



SW1.

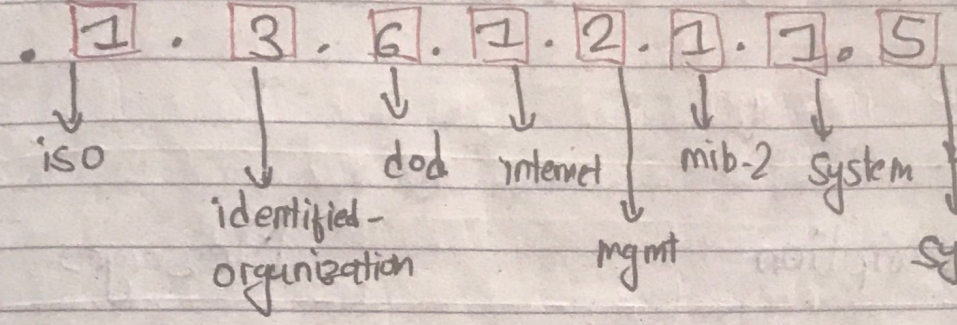
- The **SNMP Agent** is the SNMP software running on the managed devices that interacts with the SNMP Manager on the NMS.

- [redacted] Management Information Base (MIB) is the structure that contains the variables are managed by SNMP.
  - ↳ Each variable is identified with an Object ID (OID).

SNMP OIDS

SNMP OIDs are organized in a hierarchical structure.

Eg:



- There are countless different OIDs. (use oid-info.com)

SNMP Versions.

- Many versions of SNMP have been purposed/developed, however only three major versions have achieved wide-spread use.

- SNMPv1
  1) The original version of SNMP.

- SNMPv2c.
  1) Allows the NMS to ~~fee~~ retrieve large amounts of information in a single request, so its more efficient.
  → many versions of SNMPv2, but v2c is most widely used
  2) 'c' refers to 'community string' used as passwords in SNMPv1, removed from SNMPv2, and then added back for SNMPv2c.

Scanned with CamScanner

- SNMP v3.
  Much more secure version of SNMP that supports strong encryption and authentication. Whenever possible, this version should be used.

SNMP messages.

Message Class	Description	Messages.
Read	Messages sent by NMS to read information from managed devices. (i.e What's your current CPU usage? %)	Get Get Next Get Bulk
Write.	Messages sent by NMS to change information on the managed devices. (ie. change an IP address)	Set.
Notification.	Messages sent by the managed devices to alert the NMS of a particular event. (i.e interface going down)	Trap Inform.
Response.	Messages sent in response to a previous message/request	Response.

SNMP Read Messages

- Get - A request sent from the manager to the agent to retrieve the value of a variable (OID) or multiple variables. The agent will send a Response message with current value of each variable.
- Get Next
  - Request sent from manager to the agent to discover the available variables in MIB.
- GetBulk.
  A more efficient version of the GetNext message (introduced in SNMP v2)

SNMP Sent Message.

- Set.
  - These are sent by NMS to change information on managed devices. A req from manager to agent to change the value of one or more variables. The agent will send a Response message with new values

Notification Message)

- Sent by managed devices to alert the NMS of particular event.
- TRAP. TRAP
  A notification sent from the agent to the manager. The manager doesn't send a Response message to acknowledge that it received the SNMP Trap, so these msgs are unreliable.
- Inform
  A notification mssg that is acknowledged with a Response message. Originally used for communication between manager, but later updates allows agents to send inform mssg to managers, too

Response message.

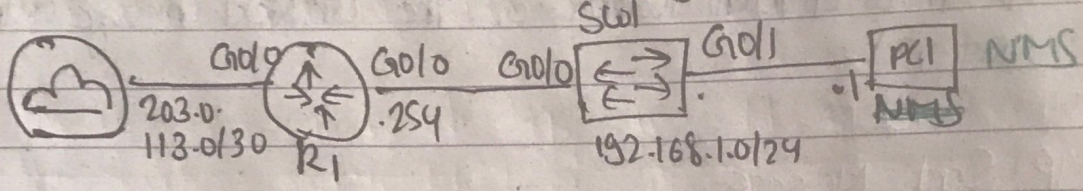
- Response – mssg sent in response to previous mssg or (repeated) request. no read

SNMP Agents = UDP.

SNMP Agent - UDP 161
SNMP Manager - UDP 162

SNMP v2c Configuration

- No NMS setup, just SNMP agent for CCNA.



- R1(config)# snmp-server contact jeremy@jeremysitlab.com
- R1(config)# snmp-server location Jeremy's House.
  - These two were optional info

- R1(config)# snmp-server community Jeremy1 ro
- R1(config)# snmp-server community Jeremy2 rw
  - Configure the SNMP community strings ~~(SNMP foo~~ (passwords)

Public: - ro - read only - no set mssg
private - rw - read/write - can use set mssgs

- R1(config)# snmp-server host 192.168.1.1 version 2c Jeremy1
  & Specify NMS, version & community

(Repeated) R1(config) # snmp-server host 192.168.1.1 version 2c Jeremy7.
- no read

1. R1(config)# snmp-server enable traps snmp linkdown linkup
R1(config)# snmp-server enable traps config.
Configure the traps types to send to NMS

In SNMPv1 & SNMPv2c, there's no encryption. The community & mssg contents are sent in plain text. This isn't secure as the packets can easily be captured & read.

SNMP config isn't mentioned in exam topic
same for syslog